

PKCERT AI & Software Development Internship

Task 02 – Python Fundamentals & Introduction to AI/ML

Objective: Strengthen Python programming fundamentals and develop a basic understanding of AI and Machine Learning concepts.

Total Marks: 100

Part A – Python Fundamentals (30 Marks)

1. Demonstrate Python data types and explain their uses.
2. Create user-defined functions with parameters and return values.
3. Develop examples using list and dictionary comprehensions and explain their benefits.
4. Create an OOP application demonstrating classes, objects, constructors, and inheritance.

Part B – Python for Data Handling (25 Marks)

1. Read, write, and append data to text files.
2. Demonstrate exception handling.
3. Read and display data from CSV and JSON files.

Part C – Development Environment (10 Marks)

1. Prepare a one-page report on Virtual Environments, Conda Environments, and dependency isolation.

Part D – Introduction to AI & Machine Learning (25 Marks)

1. Define Artificial Intelligence.
2. Differentiate AI, Machine Learning, and Deep Learning.
3. Explain supervised, unsupervised, and reinforcement learning with examples.
4. Describe the Machine Learning pipeline.

Part E – AI Libraries Research (10 Marks)

1. Prepare a comparison table for NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch showing purpose and applications.

Evaluation Criteria

Python Fundamentals

30

File Handling & Data Processing	25
Development Environment	10
AI & Machine Learning Concepts	25
AI Libraries Research	10
Total	100